Topic 1: Working as a Physicist

Throughout their study of physics at this level, students should develop their knowledge and understanding of what it means to work scientifically. They should also develop their competence in manipulating quantities and their units, including making estimates.

Students should gain experience of a wide variety of practical work that gives them opportunities to develop their practical and investigative skills by planning, carrying out and evaluating experiments. Through studying a range of examples, contexts and applications of physics, students should become increasingly knowledgeable of the ways in which the scientific community and society as a whole use scientific ideas and methods, and how the professional scientific community functions.

Students should develop their ability to communicate their knowledge and understanding of physics in ways that are appropriate to the content and to the audience.

It is not intended that this part of the specification be taught as a discrete topic. Rather, the knowledge and skills specified here should pervade the entire course and should be taught using examples and applications from the rest of the specification.

Students should:	
1.	know and understand the distinction between base and derived quantities and their SI units
2.	be able to demonstrate their knowledge of practical skills and techniques for both familiar and unfamiliar experiments
3.	be able to estimate values for physical quantities and use their estimate to solve problems
4.	understand the limitations of physical measurement and apply these limitations to practical situations
5.	be able to communicate information and ideas in appropriate ways using appropriate terminology
6.	understand applications and implications of science and evaluate their associated benefits and risks
7.	understand the role of the scientific community in validating new knowledge and ensuring integrity
8.	understand the ways in which society uses science to inform decision making